

A PROJECT TITLE

ON

Customer Credit Scoring Using XGBoost and Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATIONAL TECHNOLOGY

Name of Title: Customer Credit Scoring Using XGBoost and Logistic Regression

1. Abstract:

Customer credit scoring is a critical task in the banking and financial industry, enabling institutions to evaluate the creditworthiness of loan applicants and reduce the risk of financial losses caused by loan defaults. Traditional credit assessment methods often rely on manual evaluation and fixed rule-based systems, which may fail to capture complex relationships among customer attributes. This project proposes a machine learning-based credit scoring system that combines **Logistic Regression** and **Extreme Gradient Boosting (XGBoost)** to improve the accuracy and reliability of credit risk prediction.

The proposed system utilizes historical customer data containing demographic, financial, and behavioral attributes such as age, income, employment status, loan amount, repayment history, credit utilization, and existing debts. Before model training, the dataset undergoes preprocessing steps including data cleaning, handling missing values, encoding categorical variables, feature scaling, and feature selection to improve model performance. Logistic Regression serves as a baseline classification model due to its simplicity, interpretability, and ability to estimate the probability of default. XGBoost, an advanced ensemble learning algorithm, is employed to capture complex nonlinear relationships and interactions between features, resulting in improved predictive performance.

The trained models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score to determine their effectiveness in distinguishing between low-risk and high-risk customers. Comparative analysis is conducted to assess the strengths and limitations of both algorithms. While Logistic Regression provides easily interpretable results suitable for regulatory compliance, XGBoost generally achieves higher prediction accuracy and better generalization through gradient boosting and regularization techniques.

The proposed credit scoring system assists financial institutions in making faster, more accurate, and data-driven lending decisions. By identifying potential defaulters before loan approval, the system minimizes credit risk, reduces financial losses, and enhances operational efficiency. Furthermore, the implementation of machine learning techniques supports fair and consistent credit evaluation while enabling organizations to adapt to evolving customer behavior and market conditions. Overall, this project demonstrates the effectiveness of combining traditional statistical methods with modern machine learning algorithms to build a robust and reliable customer credit scoring solution.

2. INTRODUCTION:

Credit scoring is an essential process used by banks and financial institutions to evaluate a customer's ability to repay loans and minimize the risk of defaults. Traditional credit assessment methods are often time-consuming and may not accurately capture complex patterns in customer data. With the growth of data and advancements in technology, machine learning has become an effective approach for improving the accuracy and speed of credit risk prediction.

This project develops a **Customer Credit Scoring System** using **Logistic Regression** and **XGBoost**, two widely used machine learning algorithms. Logistic Regression provides a simple and interpretable baseline model, while XGBoost offers higher predictive accuracy by capturing complex relationships within the data. The system analyzes customer information such as age, income, employment status, credit history, loan amount, and repayment behavior to classify applicants as low-risk or high-risk.

Before training the models, the dataset is preprocessed by handling missing values, encoding categorical variables, and selecting relevant features. The performance of both models is evaluated using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. The proposed system enables financial institutions to make faster, more reliable, and data-driven lending decisions, reducing credit risk while improving overall efficiency and fairness in the loan approval process.

3. SOFTWARE REQUIREMENTS:

Software	Purpose
Operating System	Windows 10/11
Programming Language	Python 3.x
IDE	Jupyter Notebook / Google Colab / VS Code
Machine Learning Library	Scikit-learn
Gradient Boosting Library	XGBoost
Data Analysis Library	Pandas
Numerical Computing Library	NumPy
Data Visualization	Matplotlib, Seaborn
Dataset Format	CSV File

Required Python Libraries

- Python 3.x
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Matplotlib
- Seaborn

Signature of the Guide

Course Instructor